Bachelor Degree Project

Bankruptcy Prediction in the Swedish Hospitality Sector After Covid-19

- A Comparative Analysis of Altman's Z''-Score and Logistic Regression



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Abstract

This study examines how the Altman Z”-score and logistic regression generate related and complementary insights into bankruptcy risk among Swedish hospitality firms in the post Covid-19 period. The hospitality sector was particularly exposed to financial pressure during and after the pandemic due to demand volatility, high fixed costs and prolonged uncertainty, raising concerns about the continued applicability of established bankruptcy prediction models in a post crisis environment.

Using a quantitative and deductive approach, the study analyses firm level financial data for Swedish hospitality firms over the period 2017-2019 and 2021-2023, excluding 2020 due to exceptional pandemic related distortions. Both the models are applied to the same firm year observations using the same accounting based financial ratios representing liquidity, retained earnings, profitability and solvency. This design allows for a controlled comparison in which differences in results reflect model structure rather than variation in input variables.

The results show that the models capture similar underlying patterns of financial distress but differ in how this risk is represented. The Altman Z”-score provides a transparent classification framework based on predefined thresholds, while logistic regression produces probabilistic estimates and enables variable level interpretation. Liquidity and retained earnings emerge as the most significant predictors of bankruptcy risk, whereas profitability and solvency are not statistically significant within the sample. Although both models broadly identify the same distress signals, differences in classification outcomes indicate that each model captures financial risk through distinct analytical mechanisms.

The study contributes by demonstrating that traditional bankruptcy prediction models provide complementary rather than competing insights when applied to the same financial data in a sector specific and post crisis context. This highlights the importance of context dependent model evaluation, particularly in service oriented industries exposed to structural economic disruption.

Keywords: Bankruptcy prediction, Altman Z”-score, logistic regression, financial distress, hospitality sector, Covid-19, Sweden.

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1. Introduction

Business activity inherently involves financial risk. While some firms achieve stability and long term growth, others experience financial difficulties that may ultimately lead to bankruptcy. Identifying such vulnerabilities at an early stage is therefore important for business owners, lenders, investors and policymakers seeking to reduce losses and support economic stability. Bankruptcy prediction addresses this challenge by using financial information to assess a firm's likelihood of failure (Shi & Li, 2019).

Financial distress refers to persistent difficulties in meeting financial obligations, which if unresolved, may result in bankruptcy which is a formal legal process triggered by insolvency (Altman & Hotchkiss, 2006).

The later adaptation of the Altman Z-score, the Z"-score extends applicability to non manufacturing and service oriented firms (Altman, 2000). In contrast, logistic regression estimates bankruptcy probability directly from financial variables, allowing risk to be expressed in probabilistic terms (Ohlson, 1980; Succurro, 2017).

However the effectiveness of bankruptcy prediction models is context dependent. Models developed in specific industries or economic conditions may perform differently when applied elsewhere. This highlights the importance of evaluating prediction tools within defined sectoral and temporal contexts.

1.1 Background

The Swedish hospitality sector includes hotels, restaurants, cafés, bars and selected tourism related service businesses. These firms depend heavily on tourism, consumer mobility and discretionary spending, while typically operating with relatively low profit margins and high fixed costs. As a result even moderate declines in revenue can generate significant financial pressure and increase vulnerability to financial distress (Chen, Kim & Kim, 2005).

The Covid-19 pandemic constituted a major economic shock for this sector. Mobility restrictions, changes in consumer behavior and temporary closures caused abrupt revenue losses across hospitality firms (Ide, 2021). In Sweden the crisis created substantial uncertainty and increased pressure on business viability, despite comparatively less restrictive containment measures (Andersen et al., 2022; Eib & Bernhardoettel, 2023).

Policy interventions, including tax deferrals, lending programmes, and monetary easing, helped alleviate short term liquidity constraints. However evidence suggested that these measures primarily addressed immediate pressures rather than underlying financial vulnerabilities (Tascón et al., 2024).

The recovery period has been uneven, with many firms continuing to face uncertain demand and elevated leverage. Unlike typical economic downturns, the Covid-19 shock combined demand disruption with extensive policy intervention, potentially altering the financial relationships underlying bankruptcy prediction models (Crespí-Cladera et al., 2021; Tascón et al., 2024).

1.2 Bankruptcy Prediction Models and Sectoral Challenges

Bankruptcy prediction models translate financial statement data into indicators of financial risk. Ratio-based models such as the Altman Z-score provide structured insights into liquidity, profitability, solvency (Altman, 1968). The Z”-score extends this framework to non manufacturing firms, improving its applicability to service based sectors (Altman, 2000).

Despite its continued relevance, the Altman framework was originally developed using manufacturing data, which limits direct applicability to service oriented industries. Differences in asset structures, cost dynamics and revenue generation may affect the behaviour of financial ratios across sectors (Alareeni & Branson, 2012; Altman & Hotchkiss, 2006). In addition, the use of fixed coefficients may reduce reliability during periods of structural change, when historical relationships weaken (Shumway, 2001).

Logistic Regression, on the other hand, allows the contribution of individual ratios to be assessed and offers greater flexibility in modelling changing financial conditions (Ohlson, 1980). While the Z”-score produces a composite classification, logistic regression provides probabilistic estimates and variable level interpretation.

These differences suggest that the two approaches process the same financial information in distinct ways. This distinction is analytical rather than informational, as both models rely on identical financial ratios but transform them into different representations of risk. Their combined application may therefore provide a more comprehensive view of financial distress, particularly in service based sectors and post crisis environments where financial dynamics may diverge from historical patterns (Altman et al., 2014; Bellovary et al., 2007).

1.3 Problem Formulation, Motivation and Expected Contribution

The financial environment facing Swedish hospitality firms after the Covid-19 pandemic differs significantly from the conditions under which many bankruptcy prediction models were originally developed. Revenue disruptions, uneven recovery and increased leverage have altered firms' financial structures, particularly in sectors characterized by high fixed costs and demand sensitivity (Crespí-Cladera et al., 2021; Diakomihalis, 2012).

These changes challenge the assumptions underlying both ratio based and statistical prediction models, which typically rely on relatively stable relationships between accounting variables and financial distress (Altman & Hotchkiss, 2006; Shumway, 2001). As a result, there is increased uncertainty regarding whether established models continue to provide reliable distress signals in the post crisis hospitality context.

By focusing on Swedish hospitality firms in a post pandemic setting, the study contributes to understanding how model outputs relate to each other and how different modelling approaches interpret financial distress under changing economic conditions.

1.4 Scope/Limitation

This study focuses on Swedish hospitality firms, including hotels, restaurants, cafés, bars, and selected leisure or tourism related businesses, over the period affected by and following the Covid-19 pandemic. The analysis relies on firm level financial statement data and applies two established approaches, the Altman Z"-score and logistic regression.

More complex modelling techniques, such as machine learning methods, are intentionally excluded to maintain interpretability and comparability between models. Data availability may limit the inclusion of firms with incomplete financial records.

1.5 Outline of the Thesis

Chapter 2 presents the theoretical framework and literature review. Chapter 3 outlines the methodology and data. Chapter 4 presents the empirical results, which are discussed in Chapter 5. Chapter 6 concludes the study and outlines directions for future research.

2. Literature Review

2.1 Introduction

Corporate financial distress and bankruptcy prediction have long been central concerns in financial economics and accounting, given the significant consequences of firm failure for creditors, employees, and the wider economy. The early identification of financial deterioration is therefore critical for assessing bankruptcy risk before formal insolvency occurs (Altman, 1968; Beaver, 1966).

Initial research in this field relied on univariate analysis of financial ratios to distinguish between failed and non failed firms. Beaver (1966) demonstrated that deterioration in liquidity, profitability and leverage is observable prior to bankruptcy, establishing accounting information as a useful early warning signal. However subsequent studies highlight a fundamental limitation of this approach, financial distress rarely arises from a single dimension of firm performance. Instead, it reflects interacting financial pressures that produce ambiguous and often conflicting signals when analysed in isolation (Altman, 1968).

As a consequence, bankruptcy prediction research shifted toward multivariate models that combine multiple financial indicators into a unified analytical framework. One of the most influential contributions is Altman's Z-score, which applies Multiple Discriminant Analysis (MDA) to integrate accounting ratios into a composite measure of financial distress (Altman, 1968). At the same time, its reliance on a fixed linear structure has also been subject to sustained critique, particularly regarding its adaptability across industries and economic conditions.

Alongside discriminant based models, probability based approaches such as logistic regression have gained prominence. Ohlson (1980) introduced logistic regression by modelling failure risk as a probability rather than a categorical outcome, allowing for the simultaneous evaluation of multiple financial variables. Its weaker distributional assumption and flexibility in handling heterogeneous data have contributed to its widespread application in empirical bankruptcy research (Balcaen & Ooghe, 2006; Ohlson, 1980). Unlike composite scoring models, logistic regression enables variable level interpretation, but at the cost of reduced intuitive classification when applied in practice.

Both discriminant based and probabilistic approaches are grounded in the same underlying financial information, yet they differ in how this information is transformed into representations of bankruptcy risk (Altman, 1968; Ohlson, 1980). Although both models use similar accounting variables, they differ in how bankruptcy risk is represented and interpreted. This distinction is methodological rather than empirical, as both approaches operate on similar accounting inputs but produce structurally different outputs. As such differences in results may reflect model design as much as underlying financial conditions, a consideration that is central to comparative analysis.

The importance of examining these differences increases when bankruptcy prediction models are applied outside their original contexts. Altman (2000) noted that traditional Z-score models were calibrated using manufacturing firms, raising concerns regarding their applicability to service-oriented industries characterized by different financial structures. Adaptations such as the Z”-score were subsequently developed to improve relevance in non manufacturing settings, yet their effectiveness remains context dependent.

The hospitality sector provides a particularly relevant setting for such evaluation. Firms in this sector typically operate with high fixed costs and demand that is sensitive to exogenous factors such as tourism and consumer mobility (Gu & Kim, 2002). These structural characteristics increase vulnerability to revenue shocks and liquidity constraints. The Covid-19 pandemic intensified these vulnerabilities by combining abrupt demand contractions with persistent operating costs, thereby disrupting the financial conditions on which many prediction models are implicitly based (Gössling et al., 2020).

In the Swedish context, the pandemic unfolded under comparatively less restrictive containment measures, yet still resulted in significant financial pressure and uneven recovery across hospitality firms (Andersen et al., 2022). This combination of structural vulnerability and external disruption creates a setting in which historical relationships between financial ratios and bankruptcy risk may be weakened. Consequently the reliability of established prediction models cannot be assumed and required empirical reassessment (Shumway, 2001).

The Swedish hospitality sector therefore provides a suitable empirical context for examining how different modelling approaches capture financial distress under conditions of heightened uncertainty. Rather than seeking to identify a single superior model, the focus shifts toward understanding how alternative approaches interpret the same financial information and whether they generate consistent or complementary insights (Altman et al., 2017).

Accordingly, this literature review pursues three objectives. First, it establishes the theoretical foundations of bankruptcy prediction by clarifying the role of financial distress , accounting information and statistical classification (Beaver, 1966; Ohlson, 1980). Second, it critically reviews the development and application of the Altman framework, logistic regression and related approaches, focusing on their respective strengths and limitations (Altman, 1968; Balcaen & Ooghe, 2006). Third, it identifies a research gap concerning the relationship between model outputs in sector specific and post crisis contexts, providing the foundation for the empirical analysis undertaken in this study (Altman et al., 2017).

2.2 Theoretical Foundations of Bankruptcy Prediction

2.2.1 Financial Distress and Bankruptcy

A central distinction in bankruptcy prediction research is between financial distress and formal bankruptcy. Financial distress refers to a condition in which a firm experiences persistent difficulty in meeting its financial obligations, typically due to declining profitability, weak liquidity or excessive leverage (Platt & Platt, 2002; Wruck, 1990). Bankruptcy, in contrast, represents a formal legal outcome and the final stage of this process rather than its origin (Altman, 1968).

Empirical research shows that bankruptcy is rarely a sudden event, but rather the result of gradual financial deterioration (Beaver, 1966; Wruck, 1990). Persistent losses reduce retained earnings and weaken equity, while rising leverage and declining liquidity constrain a firm's ability to absorb shocks (Altman, 1968). However this progression is not deterministic. Firms may respond through restructuring, cost adjustments or refinancing, meaning that similar financial conditions do not necessarily result in bankruptcy (Wruck, 1990).

Macroeconomics shocks can accelerate firm failure even when financial conditions would not imply insolvency under stable conditions (Shumway, 2001). As a result bankruptcy risk evolves cumulatively and is shaped by both firm specific characteristics and external factors.

These dynamics support a probabilistic interpretation of bankruptcy prediction, where financial distress signals indicate increased likelihood rather than certainty of failure (Beaver, 1966; Ohlson, 1980). This perspective becomes particularly important in periods of economic disruption, where historical financial relationships may weaken, reducing the reliability of models calibrated under stable conditions (Altman et al., 2017; Shumway, 2001).

2.2.2 Accounting Information and the Going-Concern Assumption

The use of accounting information in bankruptcy prediction is grounded in the going concern assumption underlying financial reporting. Financial statements are prepared on the premise that a firm will continue operating in the foreseeable future, shaping the and recognition measurement of financial data (IASB, 2018). When this assumption becomes uncertain, financial deterioration is reflected in observable changes in accounting ratios (Altman, 1968; Beaver, 1966). These indicators provide a structured representation of financial condition and form the basis of accounting based prediction models.

Despite their widespread use, accounting based measures have limitations. They are inherently backward looking and may be influenced by accounting choices, which can reduce their predictive precision. And they rely on the assumption that distressed and non distressed firms exhibit systematic differences across combinations of financial indicators (Altman, 1968; Ohlson, 1980).

This assumption supports the use of multivariate models. By combining several financial ratios, the models provide a broader view of financial distress. This is particularly relevant given that financial distress typically emerges from the interaction of liquidity constraints, declining profitability and increasing leverage rather than a single isolated weakness (Altman, 1968; Balcaen & Ooghe, 2006).

At the same time, the effectiveness of accounting based signals may be reduced in changing economic environments. Structural disruptions can alter the relationship between financial ratios and bankruptcy outcomes, raising concerns about the stability of such models over time (Shumway, 2001).

2.2.3 Classification Logic and Prediction Models

From a methodological perspective, bankruptcy prediction can be viewed as a classification problem in which firms are assigned either to risk categories or to probabilities of failure based on observable financial characteristics (Ohlson, 1980). Different approaches address this task by transforming accounting information into measures of risk through classification rules or probabilistic estimation (Altman, 1968; Balcaen & Ooghe, 2006).

Multiple Discriminant Analysis (MDA) represents one of earliest multivariate classification methods. Altman (1968) used MDA to construct a linear combination of financial ratios that maximises separation between bankrupt and non bankrupt firms. The resulting score provides a standardized classification rule and concise summary of financial condition (Altman, 1968; Altman, 2000). However MDA relies on restrictive assumptions, including multivariate normality and equal covariance structures, which are often violated in empirical datasets. This can reduce classification reliability, particularly in heterogeneous samples or during periods of financial instability (Balcaen & Ooghe, 2006; Shumway, 2001).

Logistic regression offers an alternative by modelling bankruptcy as a probability rather than a discrete outcome (Ohlson, 1980). It estimates the relationship between financial variables and failure likelihood, allowing for more flexible interpretation of risk. Unlike discriminant analysis, logistic regression does not require strict distributional assumptions, which enhances its applicability in diverse and changing environments (Balcaen & Ooghe, 2006; Shumway, 2001).

Logistic regression does not provide an inherent classification threshold, requiring externally defined cutoffs to assign observations to categories. This reduces interpretability relative to scoring models, particularly in applied contexts.

Overall, the models approach bankruptcy risk in different ways. Discriminant models emphasise classification and interpretability, whereas logistic regression prioritises flexibility and probabilistic insight. Consequently differences in empirical results may arise from model structure as much as from underlying financial conditions.

2.3 Evolution of Bankruptcy Prediction Research

2.3.1 Early Univariate Approaches

Early bankruptcy prediction research relied on univariate analysis of financial ratios to distinguish between failed and non failed firms. Beaver (1966) showed that deterioration in liquidity, profitability and leverage ratios often can be observed several years prior to bankruptcy, establishing accounting information as early warning signals of financial distress.

Despite their importance, univariate approaches proved limited in practice. Financial distress rarely arises from weakness in a single dimension, as firms may exhibit declining profitability while maintaining liquidity, or sustain high leverage without immediate failure if cash flows remain sufficient. As a result, individual ratios often generate ambiguous or conflicting signals when analysed in isolation (Altman, 1968).

2.3.2 Shift Toward Multivariate Models

Recognition of these limitations led to the development of multivariate bankruptcy prediction models. Instead of relying on a single financial ratio, later research viewed bankruptcy as the result of several interacting financial problems related to liquidity, profitability, leverage and operational efficiency (Altman, 1968). By combining multiple financial indicators, multivariate models provide a broader assessment of financial distress (Balcaen & Ooghe, 2006).

As a result multivariate approaches generally outperform univariate models, particularly in complex or unstable environments (Balcaen & Ooghe, 2006). The development of multivariate models established two principal approaches, discriminant based and probability based methods. Discriminant models, exemplified by Altman's Z-score, apply Multiple Discriminant Analysis (MDA) to construct a linear combination of financial ratios that maximises separation between failed and non failed firms (Altman, 1968). Logistic regression, in contrast, estimates bankruptcy probability directly from financial variables, providing a probabilistic representation of distress risk (Ohlson, 1980).

These approaches differ not only in technique but in how financial distress is represented. Discriminant analysis emphasises classification through a composite financial profile, whereas logistic regression focuses on estimating the likelihood of failure and enables variable level interpretation. Consequently differences in empirical outcomes may reflect model structure as much as underlying financial conditions.

2.4 Altman's Z-score

2.4.1 The original Altman Z-score

Altman's Z-score combines accounting based financial ratios into a composite indicator used to classify firms by bankruptcy risk (Altman, 1968). Developed using Multiple Discriminant Analysis (MDA), the model reflects the view that financial distress arises from interacting weaknesses in liquidity, profitability, leverage and efficiency, providing a structured measure of Overall, financial condition (Altman, 1968; Altman, 2000).

However the use of MDA introduces important methodological assumptions, including multivariate normality and equal covariance matrices across groups. These assumptions are often violated in empirical financial data, particularly in heterogeneous or crisis affected samples, which may affect classification accuracy (Balcaen & Ooghe, 2006; Shumway, 2001).

The original model was developed for publicly traded manufacturing firms, while later adaptations, most notably the Z"-score, extended its applicability to non manufacturing and service oriented firms (Altman, 2000).

The original Altman is defined as:

$$Z = 1.2\frac{WC}{TA} + 1.4\frac{RE}{TA} + 3.3\frac{EBIT}{TA} + 0.6\frac{MVE}{TL} + 1.0\frac{S}{TA}$$

where WC denotes Working Capital, RE denotes Retained Earnings, EBIT denotes Earnings Before Interest and Taxes, MVE denotes the Market Value of Equity, TL denotes Total Liabilities, S denotes Sales, and TA denotes Total Assets (Altman, 1968).

Liquidity (WC/TA) reflects short-term solvency, retained earnings (RE/TA) indicate accumulated profitability and financial maturity, EBIT/TA measures operating performance, MVE/TL captures leverage and market based financial stability and S/TA reflects asset utilization. Firms are classified into three zones based on their Z-score, values below 1.81 indicate distress, values form a grey zone of uncertainty (Altman, 1968; Altman, 2000).

2.4.2 Critiques and Adaptations

Despite its influence, the Z-score is subject to several limitations. Its reliance on market value restricts its applicability to publicly traded firms and introduces volatility unrelated to operating performance (Altman, 1968; Altman, 2000). The sales to assets ratio is also sensitive to industry characteristics and may be less informative in service oriented sectors such as hospitality, where revenue generation is less dependent on physical asset turnover (Altman, 2000; Gu & Kim, 2002).

Furthermore, the model relies on fixed coefficients estimated under historical conditions, which may reduce its reliability during periods of economic disruption (Altman et al., 2017; Shumway, 2001).

Several adaptations have been developed to address these issues. The Z”-score replaces market based variables with accounting based measures, improving applicability to private and non manufacturing firms (Altman, 2000). It is defined as:

$$Z'' = 6.56\,\frac{WC}{TA} + 3.26\,\frac{RE}{TA} + 6.72\,\frac{EBIT}{TA} + 1.05\,\frac{BVE}{TL}$$

where BVE denotes the Book Value of Equity (Altman, 2000).

While the Z”-score improves applicability, it retains the original model structure and associated limitations. In particular, it continues to rely on fixed coefficients and the assumptions underlying discriminant analysis. As a result, it may remain sensitive to structural changes in financial relationships. In this study, the Z”-score serves as a transparent benchmark for assessing financial conditions within the Swedish hospitality sector.

2.5 Logistic Regression in Bankruptcy Prediction

Introduced by Ohlson (1980), it estimates the probability of firm failure directly from financial variables rather than assigning firms to discrete categories. Its flexibility and weaker assumptions, particularly the absence of requirements for multivariate normality and equal covariance matrices, have supported its widespread adoption in empirical studies (Balcaen & Ooghe, 2006; Ohlson, 1980).

Logistic regression allows identification of which dimensions such as liquidity, profitability or leverage are most strongly associated with financial distress (Ohlson, 1980). Unlike discriminant analysis, it does not rely on restrictive distributional assumptions often violated in financial data and can be more robust in heterogeneous samples (Shumway, 2001).

However, logistic regression also has practical limitations. While it produces probability estimates, these do not provide inherent classification thresholds. Distinguishing between “high” and “low” risk therefore requires externally defined cutoff values, reducing interpretability compared to composite scoring models such as the Z-score (Balcaen & Ooghe, 2006).

In service based firms such as hospitality, this flexibility is particularly relevant. Firms operate with high fixed costs and volatile revenues, making them sensitive to liquidity pressures even when long-term profitability appears stable (Gu & Kim, 2002; Kim & Gu, 2004). Logistic regression can capture these dynamics by allowing different financial indicators to influence bankruptcy probability, providing a more nuanced assessment of emerging risk (Ohlson, 1980).

2.6 Comparative Empirical Evidence, Context, and Research Gap

Previous research shows that bankruptcy prediction models perform differently depending on industry, firm characteristics and economic conditions (Balcaen & Ooghe, 2006). Models developed for manufacturing firms may therefore be less suitable for service industries or periods of economic uncertainty, where financial conditions differ (Shumway, 2001).

Differences are also evident across modelling approaches. Discriminant models such as the Altman Z-score tend to perform well in environments similar to their estimation sample, whereas logistic regression is generally more adaptable in heterogeneous or volatile settings where financial relationships may change (Altman, 1968; Balcaen & Ooghe, 2006; Shumway, 2001). This suggests that variation in outcomes may reflect model structure as much as underlying financial information.

The hospitality sector provides a relevant context for examining these differences. Firms operate with high fixed costs and volatile revenues driven by tourism demand, mobility and seasonality, increasing exposure to liquidity constraints and operating losses (Gu & Kim, 2002; Kim & Gu, 2004). The Covid-19 pandemic intensified these pressures, as demand collapsed while fixed costs persisted (Gössling et al., 2020).

In Sweden, the sector experienced substantial revenue declines, heightened liquidity pressure and uneven recovery despite relatively less restrictive containment measures (Andersen et al., 2022). The post Covid period environment represents an atypical setting in which established relationships between financial ratios and bankruptcy risk may be weakened (Altman et al., 2017; Shumway, 2001), making it suitable for evaluating model performance under structural disruption.

The Swedish hospitality sector thus provides a suitable context for examining model performance under heightened uncertainty. This setting allows assessment of how different modelling approaches capture financial distress under changing conditions (Altman et al., 2017; Shumway, 2001).

Research Gap:

While prior studies focus primarily on predictive accuracy, limited attention has been given to how different models interpret the same financial information within a common setting. It remains unclear whether score based classification and probability based estimation produce consistent or distinct signals of bankruptcy risk in post crisis environments.

2.7 Research Question and Hypothesis

To address this gap, this study examines how the Altman Z"-score and logistic regression relate when applied to the same dataset of Swedish hospitality firms after the Covid-19 pandemic.

Research Question:

How do the Altman Z"-score and logistic regression provide related and complementary insights into bankruptcy risk among Swedish hospitality firms after the Covid-19 pandemic?

To answer this question, the following hypotheses are tested:

- H_{01}***:*** *The Altman Z"-score is not systematically related to bankruptcy risk estimates produced by logistic regression*
- H_{11}*: The Altman Z"-score is systematically related to bankruptcy risk estimates produced by logistic regression*
- H_{02}*: Logistic regression does not provide additional interpretive insight into bankruptcy risk beyond that captured by the Altman Z"-score*
- H_{12}*: Logistic regression provides additional interpretive insight into bankruptcy risk beyond that captured by the Altman Z"-score*

3. Method

3.1 Research Philosophy and Methodological Approach

This study adopts a quantitative and deductive approach, using firm level accounting data and legally registered bankruptcy outcomes to examine bankruptcy risk. A deductive approach is appropriate as the analysis applies established theoretical models to evaluate previously identified financial relationships to evaluate whether these financial relationships are still visible in the post pandemic hospitality sector.

The analysis applies two established models: the Altman Z”-score and logistic regression. The Z”-score is selected due to its adaptation for non manufacturing firms, making it suitable for service oriented hospitality firms (Altman, 2000). Logistic regression is employed because the dependent variable is binary and the model estimates bankruptcy probability directly from financial variables (Balcaen & Ooghe, 2006; Ohlson, 1980). Applying both models to identical firm year observations enables a controlled comparative design, where differences in outputs can be attributed to model structure rather than variation in inputs. The empirical analysis is conducted using IBM SPSS, where both the Logistic regression and the Multiple Discriminant Analysis models are estimated.

3.2 Research Design

This study adopts a quantitative, non experimental research design based on secondary firm level data. It examines the application of the Altman Z”-score and logistic regression to Swedish hospitality firms in the post Covid-19 period. The sample includes hotels, restaurants, cafés, bars, and selected leisure or tourism related firms, reflecting the broader scope of the hospitality sector.

The Altman Z”-score is applied as an accounting based classification model, while logistic regression is used as a probability based approach. Both models are estimated using identical firm year observations and the same financial ratios, ensuring comparability of outputs. The analysis evaluates whether the models produce related or distinct distress signals and whether logistic regression provides additional interpretive insights beyond the composite Z”-score. This design ensures that any observed differences in results can be attributed to differences in model structure rather than differences in variable selection, allowing for a direct comparison of how financial information is transformed into measures of bankruptcy risk.

3.3 Data and Sample Selection

The empirical analysis uses secondary firm level financial data for Swedish hospitality firms. Active firms are selected using industry classification such as Hotels, Motels & Cruise Lines, Restaurants & Bars, and Leisure & Recreation, while firms with unclear sectoral relevance are screened before inclusion.

The study covers 2017-2023, with 2017-2019 representing the pre-pandemic period and 2021-2023 the post Covid-19 phase. The year 2020 is excluded due to the exceptional disruption caused by the pandemic, including abrupt demand contractions, temporary policy interventions and unusually unstable which may have distorted the relationship between financial ratios and bankruptcy risk (Altman et al., 2017; Shumway, 2001). Although policy support persisted into 2021, its intensity declined and firm performance increasingly reflected underlying financial conditions, making the period suitable for analysing post crisis adjustment. Using multiple years in both periods reduces sensitivity to short term fluctuations and captures more stale financial patterns.

Financial data are obtained from firm level financial statements and financial databases, while bankruptcy data are sourced from CreditSafe Sweden based on formally registered events. Bankrupt firms are coded as 1 and active firms as 0, creating a binary dependent variable suitable for logistic regression (Balcaen & Ooghe, 2006; Ohlson, 1980).

Financial ratios from year *t* are used to assess subsequent bankruptcy risk. As firms may appear in multiple periods, the unit of analysis is the firm year observation.

Firms with missing values are excluded to ensure consistent application across models. In addition, the original dataset was highly imbalanced, with non bankrupt firms substantially outnumbering bankrupt firms. To reduce majority class bias and improve comparability between groups, the number of non bankrupt firms was reduced while retaining all bankrupt observations. As the sample includes a broader hospitality segment rather than only SNI 5510 hotels, findings should be interpreted within this defined Swedish hospitality context.

3.4 Operationalization of Variables

3.4.1 Dependent Variable

The dependent variable is bankruptcy/distress, measured as a binary outcome. Firms formally registered as bankrupt through CreditSafe Sweden are coded as 1, while active or surviving firms are coded as 0. This binary specification is appropriate for logistic regression and aligns with established bankruptcy prediction studies based on observed failure events (Balcaen & Ooghe, 2006; Ohlson, 1980).

3.4.2 Independent Variables

The independent variables are the four financial ratios used in the Altman Z”-score, representing liquidity, accumulated profitability, operating performance, and leverage (Altman, 1968, Altman, 2000; Beaver, 1966):

- Working Capital / Total Assets
- Retained Earnings / Total Assets
- EBIT / Total Assets
- Book Value of Equity / Total Liabilities

These ratios are combined to calculate the Altman Z”-score and included individually in the logistic regression model. This consistent specification enables a direct comparison between a composite scoring model and a probabilistic model using identical financial inputs.

3.4.3 Control Variables and Variable Structure

No additional control variables are included, as the study aims to assess complementarity within the Z”-score framework. The final specification therefore consists of a binary distress variable and the four financial ratios as independent variables.

3.5 Empirical Models and Data Analysis

3.5.1 Altman Z”-score as a Benchmark Model

The Altman Z”-score is used as the accounting based classification model, as it is adapted for non manufacturing firms and therefore suitable for the hospitality sector (Altman, 2000).

It is calculated as:

$$Z'' = 6.56X_1 + 3.26X_2 + 6.72X_3 + 1.05X_4$$

where:

X_1 = Working Capital / Total Assets

X_2 = Retained Earnings / Total Assets

X_3 = EBIT / Total Assets

X_4 = Book Value of Equity / Total Liabilities

For each firm year observation, the Z''-score is computed using these ratios. Lower scores indicate higher financial distress, while higher values indicate greater financial stability. Firms are classified using established thresholds: $Z'' < 1.25$ (Distress Zone), $1.25 \leq Z'' \leq 2.60$ (Grey Zone), and $Z'' > 2.60$ (Safe Zone) (Altman, 2000). These classifications are used as accounting based distress signals in the comparative analysis.

3.5.2 Logistic Regression Model

Logistic regression is applied given the binary dependent variable, where bankrupt firms are coded as 1 and non bankrupt firms as 0. The model estimates bankruptcy probability using the same financial ratios as the Z''-score (Balcaen & Ooghe, 2006; Ohlson, 1980):

$$P(Y=1) = 1/ (1+ e \text{ ^ } -(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4))$$

where Y =1 represents bankruptcy and X_1 to X_4 denote the financial ratios.

Estimated coefficients indicate the direction and significance of each variable's relationship with bankruptcy probability. Predicted probabilities are converted into classifications using a threshold of 0.20 rather than the conventional 0.50. A lower cut off improves sensitivity to bankruptcy cases by reducing the likelihood of misclassifying distressed firms, which is particularly relevant in bankruptcy prediction where Type II errors are generally considered more consequential than incorrectly classifying financially stable firms.

3.5.3 Comparative Evaluation

Both models are evaluated using the same firm year observations. The Z"-score produces a categorical classification, while logistic regression generates probability estimates and coefficient information.

Model performance is assessed using ROC curves and AUC to evaluate discriminatory ability, alongside confusion matrices reporting classification accuracy, false positives and false negatives. Firstly, consistency of signals is assessed by examining whether both models identify similar patterns of financial distress. Second, divergence in classification is analysed to determine whether differing model outputs highlight additional risk patterns. Third, additional interpretive insight is assessed through the probability estimates and variable level coefficients provided by logistic regression, which are not captured in the composite Z"-score.

3.6 Reliability and Validity

Reliability and validity are addressed through consistent measurement, transparent data handling and application of both models to identical firm year observations.

3.6.1 Reliability

Reliability refers to the consistency and replicability of the research process (Bryman, 2016). In this study, it is supported by clearly defined variables, documented data sources and the consistent use of the same financial ratios across all observations. The use of financial statement data and legally registered bankruptcy outcomes reduces subjectivity, while established models enhance replicability (Altman, 1968; Ohlson, 1980).

To mitigate the influence of extreme values in firm level data, variables are winsorised at the 1st and 99th percentiles. This reduces the impact of outliers on model estimation while maintaining consistency across both model applications.

3.6.2 Validity

Validity concerns whether the study captures the intended concepts and supports appropriate interpretation (Bryman, 2016). Construct validity is ensured through the use of established accounting ratios representing liquidity, retained earnings, operating performance and leverage, together with legally recorded bankruptcy as the dependent variable (Altman, 1968; Beaver, 1966; Ohlson, 1980).

Internal validity is strengthened by applying both models to identical firm year observations using the same financial inputs, allowing differences in outcomes to be attributed to model structure. External validity is more limited, as the study focuses on Swedish hospitality firms in the post Covid-19 period. The findings should therefore be interpreted within this specific context rather than generalised to other sectors or broader firm populations.

3.7 Methodological Limitations

First, the analysis is context specific, focusing on Swedish hospitality firms in the post Covid-19 period. While this enhances relevance to the selected empirical setting, it limits generalisability to other sectors, countries, or time periods, particularly as bankruptcy prediction models are sensitive to industry characteristics and economic conditions (Altman et al., 2017; Balcaen & Ooghe, 2006; Shumway, 2001).

Second, the sample includes a broad hospitality segment rather than only hotels. Although this improves sample size and reflects the wider sector, it introduces heterogeneity, as hotels, restaurants, cafés, bars and tourism related firms may differ in cost structures, asset composition and revenue dynamics. The findings should therefore be interpreted within the defined Swedish hospitality context rather than as sector specific to hotels.

Third, the study relies on secondary financial data, which constrains variable availability and requires the exclusion of firms with incomplete records. This may introduce selection bias, while survivorship bias may arise due to limited pre failure data for bankrupt firms.

Fourth, the models use a parsimonious specification based on the Altman Z”-score ratios. While this ensures consistency across models, it excludes other firm specific and macroeconomic determinants of bankruptcy risk. The analysis therefore captures core accounting based distress signals rather than the full set of explanatory factors.

Fifth, the study adopts a comparative and interpretive approach, focusing on how the models operate on the same dataset rather than identifying a universally optimal prediction method.

Finally, more complex techniques, such as machine learning models, are excluded. This reflects a deliberate trade-off between predictive performance and interpretability, aligned with the study's focus on comparing established and transparent approaches.

4. Results

4.1 Data Preparation and Dataset Structure

This section describes the preparation and structure of the dataset used to examine bankruptcy prediction in Swedish hospitality firms. The analysis covers the period 2017-2019 and 2021-2023, with 2020 excluded due to the exceptional disruption associated with the onset of the Covid-19 pandemic.

The initial dataset consisted of 790 non bankrupt firms and 62 bankrupt firms, resulting in a pronounced class imbalance. Because such imbalance can bias classification models toward the majority category, the sample was adjusted to improve the representation of financially distressed firms. Specifically the number of non bankrupt firms was reduced to 200, while all bankrupt firms were retained, resulting in a final sample of 262 firms.
The dataset is structured as an unbalanced panel in which each observation represents a firm in a given year. Firms may appear multiple times depending on the availability of financial data, making the firms year the unit of analysis. This structure allows the analysis to incorporate both cross sectional and temporal variation in financial characteristics. It also implies that classification outcomes, such as true positives, false negatives and Overall, accuracy, are interpreted at the observation level rather than at the firm level, which is consistent with the design of the empirical models.

The dataset therefore reflects both cross sectional variation between firms and temporal variation across periods, providing a structured basis for evaluating model performance.

Each observation includes a firm identifier, the year of observation, financial ratios and a binary bankruptcy indicator (1 = Bankrupt, 0 = Nonbankrupt). Financial data are obtained from LSEG, CreditSafe, AllaBolag and Bolagsfakta, and four ratios are constructed in line with the Altman Z”-score framework:

- Working Capital/Total Assets
- Retained Earnings/Total Assets
- EBIT/Total Assets
- Book Value of Equity/Total Liabilities

These variables capture key dimensions of financial condition, liquidity, accumulated profitability, operating performance and leverage, and are applied consistently across all models.

Data pre-processing is implemented to ensure consistency and reliability. Observations with missing or undefined values are excluded to avoid distortions in model estimation. In addition extreme values are addressed through winsorisation at the 1st and 99th percentiles. This reduces the influence of outliers, common in firm level financial data, while preserving the Overall, distribution and relative differences between observations.

The combination of panel structure, controlled sample composition and consistent variable construction provides a robust basis for comparing how different modelling approaches process the same financial information.

4.2 Descriptive Statistics

This section presents descriptive statistics for the financial ratios $X_1 - X_4$, comparing bankrupt and non bankrupt firms and examining variation over time. The analysis focuses on central tendencies, dispersion and distributional characteristics relevant for understanding how financial indicators differ across firm outcomes.

4.2.1 Comparison Between Bankrupt and Non Bankrupt Firms

Descriptive Statistics					
Category	Particulars| Variable	X1	X2	X3	X4
Bankrupt Firms	Mean	-7.480109001	0.5758067628	-0.2664614934	0.2375049367
	Standard Deviation	92.75118805	6.912063715	0.8469694287	1.099202506
	Minimum	-1173.352676	-4.25143	-7.022321429	-0.627691458
	Maximum	0.8932926829	86.796720 38	0.5612192623	13.49680667
Non Bankrupt Firms	Mean	-0.000372679945	0.2169595265	-0.2692479005	0.5078309471
	Standard Deviation	0.3091672235	0.2071438968	1.209696163	1.259805101
	Minimum	-1.418918919	-1.527027027	-9.848635236	-0.5714285714
	Maximum	0.8232104121	0.9428081942	1.207639569	16.59717768

Table 1 - Descriptive Statistics of Financial Ratios for Bankrupt and Non Bankrupt Firms

Table 1 summarizes the distribution of the four financial ratios for bankrupt and non bankrupt firms. The results reveal clear differences in variability and the presence of the extreme observations, rather than consistent differences in average values.

Liquidity (X_1) exhibits the most pronounced contrast. Bankrupt firms show extremely high dispersion, reflected in both large standard deviations and highly negative minimum values. This indicates substantial instability in working capital relative to assets among distressed firms. In contrast, non bankrupt firms display tightly concentrated values around zero, suggesting more stable short-term financial positions. The magnitude of dispersion among bankrupt firms also indicates the presence of outliers, consistent with the volatility associated with financial distress.

For retained earnings (X_2), differences between groups are more evident in dispersion than in mean values. Bankrupt firms show wide variation, including extreme positive and negative values, indicating heterogeneous financial histories. Non bankrupt firms, by comparison, exhibit relatively narrow distributions. This pattern suggests that variability in accumulated earnings, rather than level alone, differentiates the two groups.

Profitability (X_3) shows substantial overlap across bankrupt and non bankrupt firms. Both groups exhibit low or negative average values, while dispersion remains relatively high. This overlap indicates that profitability alone provides limited separation between groups, particularly when considered without additional financial indicators.

Solvency (X_4) shows higher average values among non bankrupt firms, reflecting stronger equity positions. However variability is also significant for both groups, reducing its ability to distinguish consistently between outcomes.

Overall, bankrupt firms are characterized by higher dispersion and a greater frequency of extreme values across all ratios. In contrast nonbankrupt firms display more stable and concentrated distributions. This suggests that variability in financial conditions is a key distinguishing feature within the dataset.

4.2.2 Time Variation and Covid-19 Effects

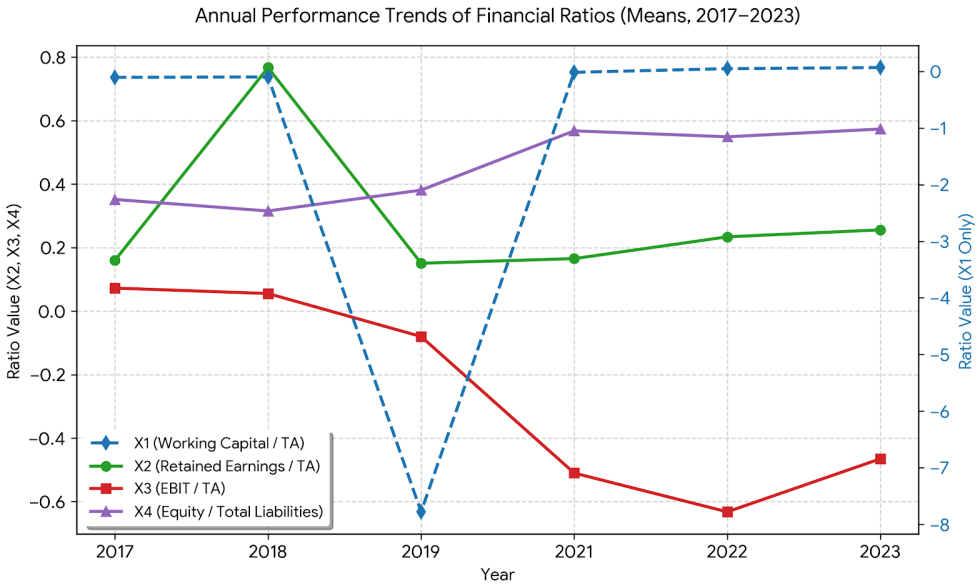


Fig. 1 - Mean Financial Ratio Trends for Swedish Hotel Firms (2017–2023)

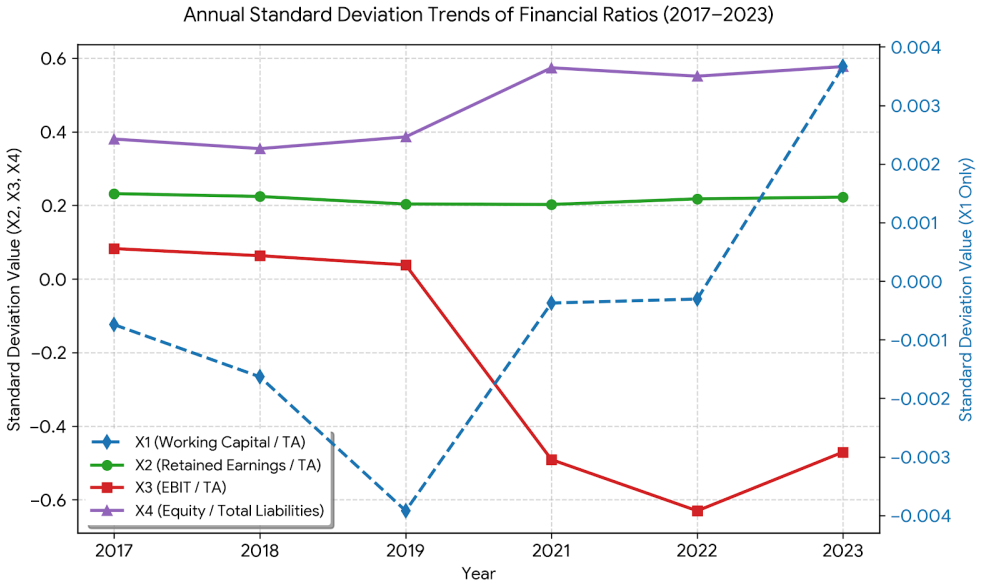


Fig 2. - Standard Deviation of Financial Ratio Trends for Swedish Hotel Firms (2017–2023)

Descriptive statistics across years indicate variation in financial indicators around the Covid-19 period. Liquidity (X_1) declines in the prepandemic period and shows partial recovery in subsequent years, although variation remains elevated. Retained earnings (X_2) exhibit higher dispersion prior to the pandemic, followed by more stable patterns in later periods.

Profitability (X_3) declines after 2019 and remains low or negative throughout the post pandemic period, indicating sustained pressure on operating performance. Solvency (X_4) remains positive on average but becomes more variable, reflecting changes in capital structure over time.

These patterns indicate increased dispersion in financial indicators around the Covid-19 period, with variability persisting into the recovery phase.

4.2.3 Correlation Analysis

	X1	*X2*	*X3*	*X4*
X1	1			
X2	0.003266261314	1		
X3	-0.007273813991	0.01805995305	1	
X4	0.01374509513	0.02480488154	0.03922436019	1

Table 2 - Correlation Matrix of Financial Ratios

Table 2 shows pairwise correlations between the financial ratios are low across all variables, with coefficients close to zero. The absence of strong correlations implies that multicollinearity is not a concern in the empirical models. As a result, the inclusion of all four ratios in subsequent analyses is unlikely to introduce redundancy or distort estimated relationships.

4.3 Model Implementation

This section outlines the implementation of the Altman Z”-score, logistic regression and Multiple Discriminant Analysis (MDA) using identical firm year observations and financial ratios $X_1 - X_4$.

4.3.1 Altman Z”-score

The Altman Z”-score is implemented as a ratio-based classification model. For each firm year observation, a composite score is calculated as a weighted linear combination of the four financial ratios. Observations are assigned to discrete risk categories based on established threshold values:

- Distress Zone: Z” < 1.25
- Grey Zone: 1.25 ≤ Z” ≤ 2.60
- Safe Zone: Z” > 2.60

The use of predefined thresholds standardizes interpretation and ensures comparability across observations, regardless of firm characteristics or time period.

Unlike statistical models, the Z”-score does not involve parameter estimation from the data. Instead it applies fixed coefficients derived from prior calibration. As a result, the model produces deterministic classifications based on observed ratio values.

4.3.2 Logistic Regression

Logistic regression is implemented using the same financial ratios with bankruptcy defined as a binary outcome (1 = bankrupt, 0 = non bankrupt). The model estimates the probability of bankruptcy as a nonlinear function of the explanatory variables:

$$P(Y=1) = 1/ (1+ e \text{ ^ } -(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4))$$

Each observation is assigned a probability value between 0 and 1 representing the estimated likelihood of bankruptcy given its financial characteristics. The estimated coefficients reflect the direction and statistical relevance of each variable’s association with the dependent variable.

In contrast to the Z”-score, logistic regression derives its parameters directly from the sample data allowing relationships between financial ratios and bankruptcy outcomes to be estimated empirically.

To enable comparison with classified models, predicted probabilities are converted into categorical outcomes using a threshold of 0.20. This threshold accounts for the class imbalance in the dataset and shifts classification toward improved detection of bankrupt observations.

4.3.3 Multiple Discriminant Analysis (MDA)

MDA is implemented as a linear classification model that constructs a discriminant function of the form:

$$D = a_1X_1 + a_2X_2 + a_3X_3 + a_4X_4$$

This function assigns each observation a discriminant score, which is used to separate bankrupt and non bankrupt observations based on a determined cut off value. The coefficients are estimated to maximise the separation between the two groups.

Unlike logistic regression, MDA does not estimate probabilities but instead focuses on creating a linear boundary that best distinguishes between groups. The effectiveness of this separation depends on the distributional properties of the data, particularly the assumption that variables are normally distributed and that covariance structures are similar across groups. Because MDA relies on these assumptions, its performance is sensitive to deviation in the data including skewness, outliers and heterogeneity in variance.

Comparative framework

The three models represent different transformations of identical financial inputs:

- Z”-score predefined rule based classification using fixed coefficients
- Logistic regression, probability based estimation with data driven coefficients and threshold classification
- MDA, linear discriminant classification based on maximizing group separation

Using the same dataset for all models makes it easier to compare how each method identifies bankruptcy risk. This structure enables a consistent and controlled comparison of how alternative methods translate financial information into measures of bankruptcy risk.

4.4 Empirical Results

This section presents the empirical findings obtained from applying the Altman Z”-score, logistic regression and Multi Discriminant Analysis (MDA) to a consistent set of firm year observations.

The analysis evaluates three core aspects of model performance, statistical significance and explanatory power, classification accuracy, error structure and discriminatory ability across observations.

4.4.1 Logistic Regression Results

Model Fit and Significance

The logistic regression model is statistically significant at the Overall, level, indicating that the included financial ratios jointly contribute to explaining bankruptcy outcomes. Despite this statistical significance, the explanatory power remains limited. The reported Nagelkerke R^2 of approximately 0.097 suggest that less than 10% of the variation in bankruptcy status is explained by the model. This relatively low pseudo R^2 indicates that while the variables contribute to distinguishing outcomes, a substantial portion of variation remains unaccounted for.

The Hosmer Lemeshow test does not reject the null hypothesis of good fit, including that predicted probabilities align reasonably with observed outcomes across probability groupings. However, this measure reflects calibration rather than classification effectiveness.

Classification Performance

	Predicted Nonbankrupt	Predicted Bankrupt
Actual Nonbankrupt	771	7
Actual Bankrupt	51	10

Table 3-Logistic regression Confusion Matrix (Threshold = 0.20)

The model achieves an Overall, classification accuracy of approximately 93%. However, this aggregate measure masks a strong asymmetry in classification performance.

The model correctly classifies approximately 99% of non bankrupt observations, indicating strong performance for the majority class. In contrast only about 16% of bankrupt observations are correctly identified, resulting in a high Type II error rate of approximately 84%. This indicates that the model frequently classifies bankrupt firms as non bankrupt.

The Type I error rate remains low (approximately 1%), indicating that few non bankrupt firms are incorrectly classified as bankrupt. This pattern reflects a conservative classification profile, in which the model prioritises minimising false positives at the expense of missing bankruptcy events. Even with a reduced threshold (0.20), the model remains strongly influenced by the distribution of the dependent variable.

Coefficient Estimates and Variable Contribution

Variable	Coefficient (B)	Significance	Interpretation
X_1 (Liquidity)	-1.611	Significant	Higher liquidity reduces probability
X_2 (Retained earnings)	-0.654	Significant	Stronger accumulated performance reduces risk
X_3 (Profitability)	0.025	Not significant	Limited contribution
X_4 (Solvency)	0.083	Not significant	Limited contribution
Constant	-2.695	Significant	Low baseline probability of bankruptcy when all variables are zero

Table 4- Logistic Regression Coefficients

Liquidity (X_1) and retained earnings (X_2) are statistically significant and exhibit negative relationships with bankruptcy probability. This indicates that higher liquidity and stronger retained earnings are associated with lower likelihood of bankruptcy.

In contrast, profitability (X_3) and solvency (X_4) do not show statistically significant effects, suggesting that these variables do not contribute meaningfully to explaining variation in bankruptcy outcomes within the sample when considered alongside the other variables.

The pattern of coefficient indicates that variation in short term financial flexibility and cumulative financial performance plays a larger role in the estimated probability structure than operating profitability or leverage alone.

4.4.2 Multiple Discriminant Analysis (MDA)

Variable Contribution and Group Separation

The MDA model constructs a discriminant function aimed at separating bankrupt and non bankrupt observation through a linear combination of the financial ratios.

The results indicate that liquidity (X_1) is the only variable contributing significantly to group discrimination, while the remaining variables have relatively small contributions. This concentration of explanatory power in a single variable limits the multidimensional separation intended by the model.

Overall, group separation is weak. This is reflected in a low canonical correlation (0.136), indicating that the discriminant function explains only a small proportion of variation between groups. Similarly, the Wilkis' Lambda value remains close to one, suggesting limited distinction in the distribution of predictor variables across the two groups.

The Box's M test is statistically significant, indicating a violation of the assumption of equal covariance matrices across groups. Since MDA relies on this assumption, its violation reduces the reliability of the estimated discriminant function and contributes to the limited separation observed in the results.

Classification Performance

	Predicted Non Bankrupt	Predicted Bankrupt
Actual Non Bankrupt	778	0
Actual Bankrupt	30	31

Table 5- MDA Confusion Matrix (Threshold = 0.20)

The model achives an Overall, classification accuracy of approximately 86%, which is lower than that of logistic regression. However, the distribution of classification outcomes differs substantially.

Approximately 51% of bankrupt observations are correctly identified, representing a significant improvement relative to logistic regression. At the same time, the model produces no false positives, resulting in a Type I error rate of zero.

Type II error rate (approximately 49%) remains substantial but is considerably lower than that observed in logistic regression. This indicates that the model improves sensitivity to bankruptcy cases, although it does not achieve strong Overall, discriminatory performance.

4.4.3 ROC Curve and Discriminatory Ability

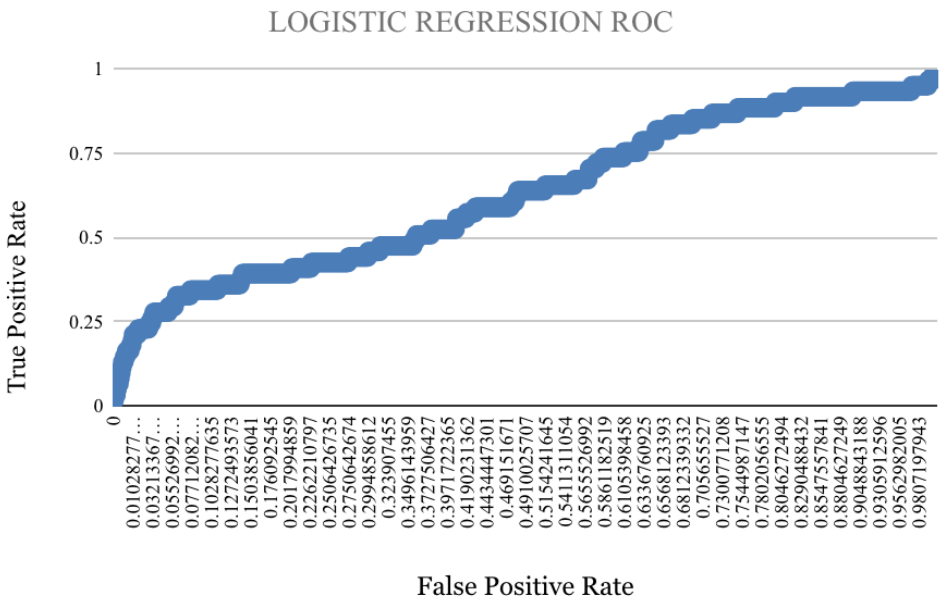


Fig. 3 - Logistic Regression ROC Curve

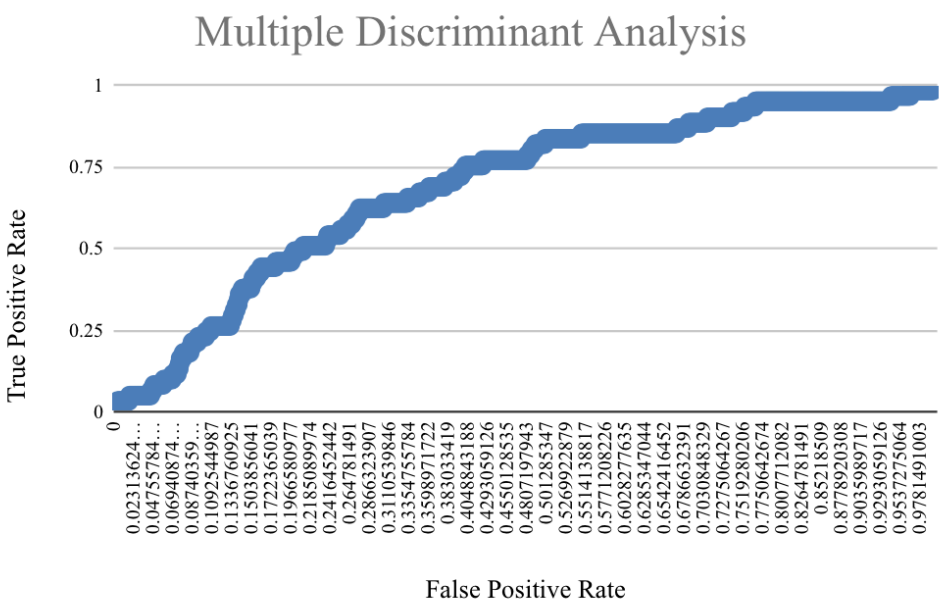


Fig. 4 - MDA ROC Curve

The Area Under the Curve (AUC) provides a summary measure of this discrimination:

- Logistic regression, AUC = 0.638
- MDA, AUC = 0.701

An AUC of 0.5 corresponds to random classification, while values above 0.7 are generally considered acceptable. Accordingly, logistic regression exhibits weak discrimination, whereas MDA reaches a modest level of performance.

The ROC curve provides additional insight beyond the AUC values. The logistic regression curve lies relatively close to the diagonal benchmark over much of its range, indicating limited improvement over random classification and weak separation between classes. In contrast the MDA curve deviates further from the diagonal, particularly in the mid range of false positive rates, indicating a stronger ability to rank observations according to bankruptcy risk.

However, neither curve approaches the upper left corner of the plot which would indicate high sensitivity combined with low false positive rates. This suggests that both models face limitations in simultaneously achieving high detection of bankrupt firms and low misclassification of non bankrupt firms.

The higher AUC for MDA indicates better ranking performance, meaning that bankrupt observations tend to receive higher discriminant scores relative to non bankrupt observations. This improvement in ranking does not fully translate into classification accuracy, as reflected in the confusion matrices.

4.4.4 Comparative Error Analysis

Measure	Logistic Regression	MDA
Overall Accuracy	93.1%	86.1%
Correctly Classified Non Bankrupt Firms	99.1%	91.3%
Correctly Classified Bankrupt Firms	16.4%	19.7%
False Positive Rate (Type I Error)	Low	Higher
False Negative Rate (Type II Error)	High	Lower
Classification Behaviour	Conservative	More sensitive to bankruptcies

Table 6 - Classification Performance of Logistic Regression and MDA

Table 6 shows a comparison of classification outcomes and highlights differences in how the models process financial information.

Logistic regression exhibits strong performance in identifying non bankrupt observations and maintains a low false positive rate. However it fails to capture most bankruptcy cases, resulting in high Type II error.

MDA in contrast, improves detection of bankruptcy cases by reducing false negatives. However this improvement does not result in superior Overall, accuracy, as classification remains incomplete and group separation is limited.

The primary difference between the models lies in sensitivity to bankruptcy events, indicating variation in how each model interprets the same financial inputs.

4.4.5 Z”-Score Classification and Model Comparison

The Z”-score assigns observations to predefined risk categories based on financial ratios. Observations classified in the distress zone contain a higher proportion of bankrupt firms, while the safe zone is associated with lower bankruptcy incidence. The gray zone contains mixed outcomes.

Across all models, liquidity consistently emerges as the most influential variable, reflecting its role in both classification and probability estimation frameworks.

4.5 Hypothesis Testing and Interpretation

This section evaluates the empirical findings in relation to the research question and hypotheses. The assessment focuses on the consistency of model outputs, the relevance of financial variables and differences in how bankruptcy risk is represented across models.

4.5.1 Hypothesis 1: Relationship between Z”-Score and Logistic Regression

H_{01}: The Altman Z”-score is not systematically related to bankruptcy risk estimates produced by logistic regression
H_{11}: The Altman Z”-score is systematically related to bankruptcy risk estimates produced by logistic regression.

Evaluation

The results show that both models capture similar patterns of financial distress using the same underlying variables. The Z”-score identifies a higher concentration of bankrupt observations within the distress category, indicating alignment between classification and observed outcomes.

Logistic regression identifies liquidity (X_1) and retained earnings (X_2) as statistically significant predictors. These variables are central components of the Z”-score, indicating that both models are driven by the same financial dimensions.

In addition, both models exhibit similar classification profiles, characterised by high accuracy for nonbankrupt observations and limited detection of bankruptcy cases.

Conclusion

H_{01} is rejected and H_{11} is supported.

The Z”-score and logistic regression produce systematically related signals of financial distress based on common financial drivers and consistent classification patterns.

4.5.2 Hypothesis 2: Complementarity of Logistic Regression

H_{02}: Logistic regression does not provide additional interpretive insight beyond the Altman Z”-score

H_{12}: Logistic regression provides additional interpretive insight beyond the Altman Z”-score.

Evaluation

The results show that logistic regression provides additional information not captured by the Z”-score framework. The Z”-score assigns observations to predefined categories, while logistic regression produces continuous probability estimates, allowing variation in risk within classification groups.

Logistic regression also provides coefficient estimates that identify the significance and direction of individual variables. This enables direct assessment of variable contribution, whereas the Z”-score aggregates inputs into a composite measure without isolating individual effects.

Finally, the use of an adjustable probability threshold introduces flexibility in classification, in contrast to the fixed structure of the Z”-score. These differences operate at the level of output structure and interpretation rather than input variables.

Conclusion

H_{02} is rejected and H_{12} is supported.

Logistic regression provides additional interpretive insight through probability estimation, variable level analysis and flexible classification.

4.5.3 Overall Interpretation

The models are consistent in the financial signals they capture but differ in representation. Both rely on the same inputs and identify liquidity and retained earnings as key drivers, but the Z”-score expresses risk categorically, whereas logistic regression represents it as a continuous probability.

4.5.4 Link to Study Objective

The findings demonstrate that bankruptcy risk is driven by common financial variables across models, while its representation depends on model structure. The results therefore confirm both alignment and complementarity in how the models process identical financial inputs.

5. Discussion

5.1 Restatement of Research Focus

This study examined how the Altman Z”-score and logistic regression provide related and complementary insights into bankruptcy risk among Swedish hospitality firms. The analysis used firm year observations to assess whether both models rely on similar financial signals and whether logistic regression offers additional interpretive value beyond a ratio based scoring model.

By applying both models to identical financial inputs, the study isolates how differences in modelling structure influence the identification and representation of financial distress.

5.2 Interpretation of Main Findings

A central finding is that both models rely on the same underlying financial signals, with liquidity and retained earnings consistently emerging as the most relevant indicators of financial distress.

At the same time, both models exhibit limited ability to identify bankruptcy events. While classification accuracy for nonbankrupt observations is high, detection of bankrupt firms remains weak. This asymmetry suggests that financial distress signals in the dataset are unevenly distributed, with stable firms displaying more consistent financial patterns than distressed firms. As a result, models identify financial stability more reliably than failure outcomes. This may indicate that bankruptcy risk in post pandemic hospitality firms is characterized by greater financial heterogeneity and instability, reducing the consistency of observable distress patterns across firms.

The results also indicate that model performance is constrained by overlap in financial characteristics between bankrupt and nonbankrupt observations. Similar ratio values can be associated with different outcomes, reducing the ability of both models to achieve clear separation. This limits classification accuracy regardless of whether a probabilistic or rule based approach is applied. This indicates that financial distress cannot be fully separated using individual accounting indicators alone, limiting the effectiveness of both classification and probability based approaches.

Despite this shared limitation, the models differ in how financial information is translated into outputs. The Z” score simplifies financial conditions into discrete risk categories, while logistic regression represents risk as a continuous probability and allows variable level interpretation. These differences do not alter the underlying signals but affect how variation in financial distress is expressed across observations.

Overall, the findings indicate that the models operate on a common informational basis but produce distinct representations of bankruptcy risk, reflecting differences in model structure rather than differences in input data.

5.3 Relation to Theory and Previous Research

The prominence of liquidity and retained earnings is consistent with the theoretical foundations of bankruptcy prediction models. Within the Altman framework, financial distress is understood as the result of interacting weaknesses in liquidity and accumulated profitability, which jointly determine a firm's ability to sustain operations (Altman, 1968; Altman, 2000). The results support this structure, as both variables consistently drive outcomes across modelling approaches, indicating that core accounting signals remain relevant in the empirical setting.

The limited ability to detect bankruptcy is also consistent with prior empirical findings. Financial ratio based models are generally more effective in identifying financially stable firms than in predicting actual failure events, as distress develops gradually and may not be fully reflected in observable accounting ratios at a single point in time (Balcaen & Ooghe, 2006). This is particularly relevant in heterogeneous or post disruption environments, where relationships between financial ratios and bankruptcy outcomes may weaken (Shumway, 2001).

The similarity in variable relevance across models supports the idea that different prediction methods often rely on the same financial information. Both discriminant based and probability based approaches transform the same underlying financial characteristics, even when their outputs differ (Altman et al., 2017; Ohlson, 1980). This supports prior research indicating that differences in model outcomes arise primarily from model structure rather than differences in underlying financial variables.

At the same time, differences in output structure confirm that model choice influences how financial information is represented. While both models capture similar signals, they impose different analytical frameworks on those signals, leading to variation in how bankruptcy risk is quantified and interpreted. This is consistent with prior criticism of static classification models, where simplified categorisation may reduce sensitivity to financial variation in unstable economic environments (Balcaen & Ooghe, 2006).

5.4 Complementarity of the Models

The findings show that logistic regression does not replace the Z”-score but instead provides a different layer of analysis. The Z”-score aggregates financial ratios into a composite classification, offering a simplified summary of financial conditions through discrete risk categories. Logistic regression, in contrast, models bankruptcy risk as a continuous probability and allows the contribution of individual variables to be evaluated within the sample.

This distinction reflects two different functions. The Z”-score provides a rule based classification that facilitates consistent and interpretable categorisation of firms, while logistic regression enables ranking and differentiation within and across these categories. As a result, observations that appear similar under categorical classification may still differ in their estimated probability of bankruptcy.

In addition, logistic regression introduces variable level interpretation through coefficient estimates, which allows the relative importance of financial ratios to be assessed directly. This provides a level of analytical detail not available in the composite structure of the Z”-score.

The complementarity between the models therefore arises not from different inputs, but from different transformations of the same financial information. The Z”-score summarises financial conditions into categorical outcomes, whereas logistic regression captures variation and relative risk within those categories.

Taken together, the models provide layered insights into bankruptcy risk: one through classification and the other through probabilistic differentiation. This combination enables a more comprehensive representation of financial distress than either approach in isolation.

5.5 Limitations of the Empirical Results

Several limitations arise from the empirical analysis and directly affect model performance.

First, classification performance is constrained by the imbalance in the dataset. Despite sampling adjustments, nonbankrupt observations remain dominant, which biases classification toward the majority category. This is most evident in logistic regression, where predicted probabilities are influenced by the underlying distribution, leading to systematically low sensitivity to bankruptcy cases.

Second, both models exhibit limited discriminatory power, as reflected in the AUC results. Even the better performing model does not achieve strong separation between bankrupt and non bankrupt observations. This indicates that the financial ratios included, while theoretically relevant, do not fully capture the variation required to distinguish outcomes consistently across firms. This may further suggest that financial distress within the post pandemic hospitality sector is influenced by additional operational and contextual factors not fully reflected in accounting based indicators alone.

Third, overlap in financial characteristics across groups further constrains prediction. This shows that similar ratio values can be associated with both bankrupt and nonbankrupt firms, indicating that accounting based indicators do not uniquely map to bankruptcy outcomes. It reduces the effectiveness of both classification and probability based approaches. The findings therefore reinforce previous research suggesting that bankruptcy prediction remains inherently probabilistic rather than deterministic.

Finally, differences in classification outcomes across models reflect the influence of model structure. Even with identical inputs, the transformation of financial information into outputs produces variation in detection and interpretation. This highlights that predictive performance depends not only on data quality but also on how models operationalise financial signals.

5.6 Contribution to the Research Objective

The findings contribute to the research objective by demonstrating that models based on the same financial ratios can produce both consistent signals and distinct representations of bankruptcy risk.
This study shows that:

- Common financial variables underpin bankruptcy risk across models
- Different modelling approaches lead to different output structures
- The representation of risk depends on how financial information is processed
- Probabilistic and classification based approaches therefore provide complementary rather than competing perspectives on financial distress.

These findings isolate model structure as the primary source of variation in outputs when financial inputs are held constant. The study therefore contributes to bankruptcy prediction research by emphasizing interpretation and model complementarity within a sector specific and post pandemic context, rather than focusing solely on predictive accuracy.

5.7 Critical Reflection

From a broader perspective, the results highlight that bankruptcy prediction is constrained by both data and model design. The models successfully identify general patterns of financial stability but are less effective in detecting failure events.

The reliance on a limited number of financial ratios restricts the ability to fully capture the complexity of financial distress. While the models reflect key aspects of firm performance, they do not incorporate all factors that may influence bankruptcy outcomes. Operational conditions, external shocks and firm specific circumstances may therefore affect bankruptcy risk in ways not fully observable through accounting based indicators alone.

In addition, the variation observed across model outputs demonstrates that no single method provides a complete representation of risk. Each model emphasises different aspects of the same financial data, and these differences affect both classification outcomes and interpretation. This may become particularly relevant during periods of economic uncertainty, where historical financial relationships are less stable and financial distress develops unevenly across firms.

The findings therefore support a more context dependent and interpretive perspective on bankruptcy prediction, where complementary approaches may provide broader insight than isolated model application.

6 Conclusion

6.1 Summary of Research Objective

This study examined how the Altman Z”-score and logistic regression generate related and complementary insights into bankruptcy risk among Swedish hospitality firms.

By applying both methods to the same firm year dataset and financial variables, the study isolates differences arising from model structure rather than data inputs. This provided a consistent basis for evaluating how bankruptcy risk is identified and represented across approaches.

6.2 Summary of Key Findings

The findings show that bankruptcy risk is consistently associated with a common set of financial indicators. In particular, liquidity and retained earnings emerge as the most influential variables across models.

At the same time, the results demonstrate that the models differ in how this information is translated into outputs. The Altman Z”-score provides a categorical classification of financial condition, while logistic regression expresses bankruptcy risk as a probability and identifies the statistical relevance of individual variables.

Another key finding is that prediction performance is uneven across models. Both approaches perform well in identifying non bankrupt observations, but neither model achieves strong detection of bankruptcy cases. This highlights a consistent limitation in the ability of financial ratio based models to identify failure events.

6.3 Answer to the Research Question

The results provide a clear answer to the research question.

The Altman Z”-score and logistic regression produce related insights because they rely on the same financial variables and consistently identify similar patterns of financial distress. Both models reflect the same underlying signals, particularly in relation to liquidity and retained earnings.

Also, the models provide complementary insights because they differ in how bankruptcy risk is represented. The Z”-score organizes financial information into predefined risk categories, while logistic regression transforms the same information into probability estimates and variable level relationships. These differences lead to distinct interpretations of risk, despite shared inputs.

The findings therefore show that the models are not substitutes, but rather provide different perspectives on the same financial reality.

6.4 Contribution of the Study

This study contributes by demonstrating how two widely used bankruptcy prediction approaches relate to each other when applied to the same dataset and financial variables. By holding inputs constant, the analysis isolates the role of model structure in shaping outputs.

The results highlight that:

- common financial variables underpin bankruptcy signals across models
- differences in modelling approaches lead to differences in interpretation
- financial ratios provide consistent but limited predictive information
- probabilistic and classification based approaches provide related but distinct representations of bankruptcy risk

These findings isolate the model structure as the main source of variation in outputs when the financial inputs are held constant. This establishes that model differences reflect analytical structure rather than informational content, providing a more precise understanding of how accounting based and statistical approaches relate in empirical bankruptcy analysis.

6.5 Limitations and Implications

The dominance of non bankrupt observations in the dataset reduces sensitivity to bankrupt cases, particularly for regression based classification. In addition, the moderate discriminatory ability observed across models indicates that financial ratios alone do not provide a complete basis for predicting firm failure. While the models identify general financial patterns, they are less effective in isolating cases of actual failure.

6.6 Suggestions for Future Research

Future research could extend this analysis in several directions. First, additional explanatory variables could be included to capture aspects of financial distress not reflected in accounting ratios, such as market based indicators or macroeconomic conditions. Second, alternative modelling approaches, including more advanced statistical or machine learning techniques, could be applied to assess whether predictive performance improves. Third, comparative studies across different industries may provide further insight into whether the observed patterns are sector specific or more broadly applicable.

6.7 Final Conclusion

Overall, the study shows that bankruptcy prediction models based on financial ratios identify consistent underlying signals, but differ in how those signals are structured and interpreted. The Altman Z”-score and logistic regression therefore provide both overlapping and complementary perspectives on financial distress.

The results show that model structure is central to translating financial information into measures of risk. Consequently, bankruptcy prediction is shaped not only by the data used, but by how that data is modelled.

AI Statement

AI tools were used in a limited way to help with data handling, such as creating Excel formulas, organising data, cleaning datasets and helping with graphs. All analysis, interpretations and written parts of the thesis were developed independently by the authors.

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Appendices

Appendix A - Logistic Regression Output

A1. Omnibus Tests of Model Coefficients and Model Summary

Omnibus Tests of Model Coefficients				
		Chi-square	df	Sig.
Step 1	Step	33.729	4	0.000
	Block	33.729	4	0.000
	Model	33.729	4	0.000

Model Summary				
Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square	
1	403.528a	0.039	0.097	

a. Estimation terminated at iteration number 10 because parameter estimates changed by less than .001.

A2. Classification Table a

Classification Tablea					
			Predicted		
			Bankruptcy		Percentage Correct
Observed			0	1	
Step 1	Bankruptcy	0	771	7	99.1
		1	51	10	16.4
	Overall Percentage				93.1

a. The cut value is .200

A3. Variables in the Equation

Variables in the Equation									
								95% C.I.for EXP(B)	
		B	S.E.	Wald	df	Sig.	Exp(B)	Lower	Upper
Step 1a	X1	-1.611	0.372	18.716	1	0.000	0.200	0.096	0.414
	X2	-0.654	0.274	5.692	1	0.017	0.520	0.304	0.890
	X3	0.025	0.110	0.053	1	0.818	1.026	0.827	1.271
	X4	0.083	0.111	0.553	1	0.457	1.086	0.874	1.350
	Constant	-2.695	0.166	262.635	1	0.000	0.068		
a. Variable(s) entered on step 1: X1, X2, X3, X4.									

Appendix B - Multiple Discriminant Ananlysis (MDA) Output

B1. Eigenvalues and Wilk's Lambda

Eigenvalues				
Function	Eigenvalue	% of Variance	Cumulative %	Canonical Correlation
1	.019a	100.0	100.0	0.136
a. First 1 canonical discriminant functions were used in the analysis.				

Wilks' Lambda				
Test of Function(s)	Wilks' Lambda	Chi-square	df	Sig.
1	0.982	15.546	4	0.004

B2. Standardized Canonical Discriminant Function Coefficients

Standardized Canonical Discriminant Function Coefficients	
	Function
	1
X1	0.919
X2	0.170
X3	0.270
X4	0.214

B3. Structure Matrix

Structure Matrix

	Function
	1
X1	0.918
X3	0.273
X4	0.241
X2	0.182

Pooled within-groups correlations between discriminating variables and standardized canonical discriminant functions
Variables ordered by absolute size of correlation within function.

B4. Classification Results a, c

Classification Resultsa,c					
			Predicted Group Membership		
Bankruptcy			0	1	Total
Original	Count	0	710	68	778
		1	49	12	61
	%	0	91.3	8.7	100.0
		1	80.3	19.7	100.0
Cross-validated db	Count	0	710	68	778
		1	50	11	61
	%	0	91.3	8.7	100.0
		1	82.0	18.0	100.0
a. 86.1% of original grouped cases correctly classified.					
b. Cross validation is done only for those cases in the analysis. In cross validation, each case is classified by the functions derived from all cases other than that case.					
c. 85.9% of cross-validated grouped cases correctly classified.					

B6. Discriminant Function

$$D = 0.712X_1 + 0.488X_2 + 0.041X_3 + 0.063X_4$$

Appendix C - Variable Definitions

Variable	Formula	Financial Dimension
X1	Working Capital/Total Assets	Liquidity
X2	Retained Earnings/Total Assets	Accumulated profitability
X3	EBIT/Total Assets	Operating performance
X4	Book Value of Equity/Total Liabilities	Solvency

Appendix D - Altman Z” - Score Formula

$$Z" = 6.56X_1 + 3.26X_2 + 6.72X_3 + 1.05X_4$$

D1. Classification Threshold

Z”-score	Interpretation
Z” < 1.25	Distress Zone
1.25 ≤ Z” ≤ 2.60	Grey Zone
Z” > 2.60	Safe Zone

D2. Example Calculation

Variable	Value
X_1	0.15
X_2	0.10
X_3	0.04
X_4	0.50

Appendix E - Data Preparation and Sample Selection

Step	Number of firms
Initial sample	852
Non bankrupt firms	790
Bankrupt firms	62
Final balanced sample	262

Appendix F - Correlation Matrix

Variables	X1	X2	X3	X4
X1	1	0.003	-0.007	0.014
X2	0.003	1	0.018	0.025
X3	-0.007	0.018	1	0.039
X4	0.014	0.025	0.039	1